which are in constant motion, any one of which can have a dramatic impact, and none of which is predictable to an absolute certainty. The task for investors, then, is to narrow the field by identifying and removing that which is most unknown, and then focusing on the least unknown. That task is an exercise in probability.

When we are unsure about the situation but still want to express our opinion, we often preface our remarks by saying "the chances are," or "probably," or "it's not very likely but. . . ." When we go one step further and attempt to quantify those general expressions, we then are dealing with probabilities. Probabilities are the mathematical language of risk.

What is the probability of a cat giving birth to a bird? Zero. What is the probability the sun will rise tomorrow? That event, which is considered certain, is given a probability of 1. All events that are neither completely certain nor completely impossible have a probability somewhere between 0 and 1, expressed as a fraction. Determining the fraction is what probability theory is all about.

In 1654, Blaise Pascal and Pierre de Fermat exchanged a series of letters that formed the basis of probability theory. Pascal, a child prodigy gifted in both mathematics and philosophy, had been challenged by the Chevalier de Méré, a philosopher and gambler, to solve the riddle that had stumped many mathematicians. How should two card players decide the stakes of the game if they had to leave before the game was completed? Pascal approached Fermat, a mathematical genius in his own right, about Méré's challenge.

"The 1654 correspondence between Pascal and Fermat on this subject," said Peter Bernstein in *Against the Gods*, his wonderfully written treatise on risk, "signaled an epochal event in the history of mathematics and the theory of probability."[8] Although they attacked the problem differently (Fermat used algebra whereas Pascal turned to geometry), they were able to construct a system to determine the probability of several possible outcomes. Indeed, Pascal's triangle of numbers solves many problems, including the probability that your favorite baseball team will win the World Series after losing the first game.

The work by Pascal and Fermat marks the beginning of the theory of decision making. Decision theory is the process of deciding what to do